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problem. We will discuss constraint handling in Bayesian optimization in Sect. 1.3.2.4.

Second, and more generally, one can apply multi-objective optimization to search for the Pareto front, a set of configurations which are optimal tradeoffs between the objectives in the sense that, for each configuration on the Pareto front, there is no other configuration which performs better for at least one and at least as well for all other objectives. The user can then choose a configuration from the Pareto front. We refer the interested reader to further literature on this topic [53, 57, 65, 134].

1.3 Blackbox Hyperparameter Optimization

In general, every blackbox optimization method can be applied to HPO. Due to the non-convex nature of the problem, global optimization algorithms are usually preferred, but some locality in the optimization process is useful in order to make progress within the few function evaluations that are usually available. We first discuss model-free blackbox HPO methods and then describe blackbox Bayesian optimization methods.

1.3.1 Model-Free Blackbox Optimization Methods

Grid search is the most basic HPO method, also known as full factorial design [110]. The user specifies a finite set of values for each hyperparameter, and grid search evaluates the Cartesian product of these sets. This suffers from the curse of dimensionality since the required number of function evaluations grows exponentially with the dimensionality of the configuration space. An additional problem of grid search is that increasing the resolution of discretization substantially increases the required number of function evaluations.

A simple alternative to grid search is random search [13].[1] As the name suggests, random search samples configurations at random until a certain budget for the search is exhausted. This works better than grid search when some hyperparameters are much more important than others (a property that holds in many cases [13, 61]). Intuitively, when run with a fixed budget of B function evaluations, the number of different values grid search can afford to evaluate for each of the N hyperparameters is only $B^{1/N}$, whereas random search will explore B different values for each; see Fig. 1.1 for an illustration.

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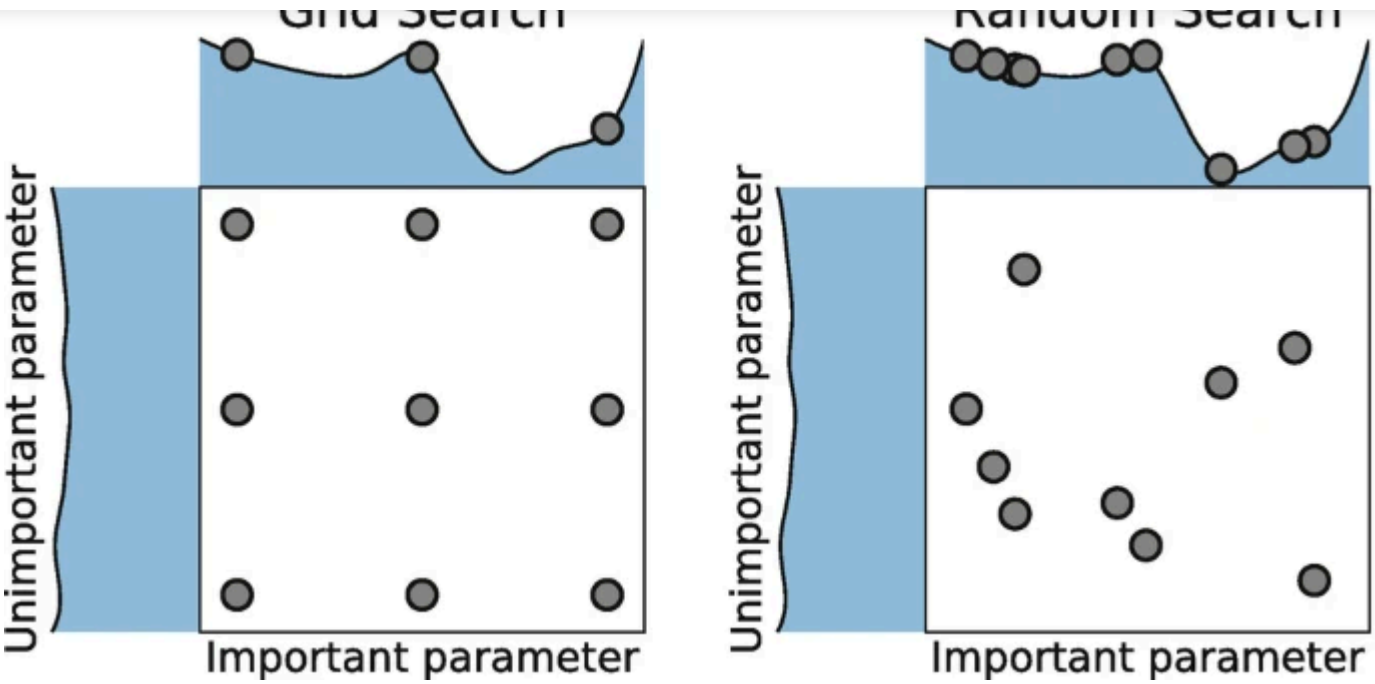

Comparison of grid search and random search for minimizing a function with one important and one unimportant parameter. This figure is based on the illustration in Fig. 1 of Bergstra and Bengio [13]

Further advantages over grid search include easier parallelization (since workers do not need to communicate with each other and failing workers do not leave holes in the design) and flexible resource allocation (since one can add an arbitrary number of random points to a random search design to still yield a random search design; the equivalent does not hold for grid search).

Random search is a useful baseline because it makes no assumptions on the machine learning algorithm being optimized, and, given enough resources, will, in expectation, achieves performance arbitrarily close to the optimum. Interleaving random search with more complex optimization strategies therefore allows to guarantee a minimal rate of convergence and also adds exploration that can improve model-based search [3, 59]. Random search is also a useful method for initializing the search process, as it explores the entire configuration space and thus often finds settings with reasonable performance. However, it is no silver bullet and often takes far longer than guided search methods to identify one of the best performing hyperparameter configurations: e.g., when sampling

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function evaluations to find the optimum, whereas a guided search could find the optimum in $N+1$ function evaluations as follows: starting from an arbitrary configuration, loop over the hyperparameters and change one at a time, keeping the resulting configuration if performance improves and reverting the change if it doesn't. Accordingly, the guided search methods we discuss in the following sections usually outperform random search [12, 14, 33, 90, 153].

Population-based methods, such as *genetic algorithms*, *evolutionary algorithms*, *evolutionary strategies*, and *particle swarm optimization* are optimization algorithms that maintain a population, i.e., a set of configurations, and improve this population by applying local perturbations (so-called mutations) and combinations of different members (so-called crossover) to obtain a new generation of better configurations. These methods are conceptually simple, can handle different data types, and are embarrassingly parallel [91] since a population of N members can be evaluated in parallel on N machines.

One of the best known population-based methods is the covariance matrix adaption evolutionary strategy (CMA-ES [51]); this simple evolutionary strategy samples configurations from a multivariate Gaussian whose mean and covariance are updated in each generation based on the success of the population's individuals. CMA-ES is one of the most competitive blackbox optimization algorithms, regularly dominating the *Black-Box Optimization Benchmarking* (BBOB) challenge [11].

For further details on population-based methods, we refer to [28, 138]; we discuss applications to hyperparameter optimization in Sect. 1.5, applications to neural architecture search in Chap. 3, and genetic programming for AutoML pipelines in Chap. 8.

1.3.2 Bayesian Optimization

Bayesian optimization is a state-of-the-art optimization framework for the global optimization of expensive blackbox functions, which recently gained traction in HPO by obtaining new state-of-the-art results in tuning deep neural networks for image classification [140, 141], speech recognition [22] and neural language modeling [105], and by demonstrating wide applicability to different problem settings. For an in-depth